Sam Prosser/Maddie McDowell/Patrick King/Thomas Powell/Nicolas Ortiz
Senior Project I

1. Plan Introduction

This software development plan outlines the approach, resources, organization, and schedule for **Pathfinder**, an all-in-one platform designed to guide users through their educational and professional journeys. The project aims to deliver a comprehensive toolset for students, graduates, and professionals, offering personalized support in career guidance, resume building, job matching, and networking, with the assistance of an AI-powered chatbot.

1.1.1 Project Deliverables

- **Platform Design Document (PDD):** A detailed document outlining Pathfinder's core features, workflows, and user interface.
- **Alpha Version:** An initial version of Pathfinder, featuring the basic career questionnaire, resume analysis, and cover letter generator tools.
- **Beta Version:** A feature-complete version of the platform, including job matching, learning resources, networking tools, and AI chatbot integration.
- **Release Version:** Final version of Pathfinder, incorporating user feedback, bug fixes, and optimization for the platform's launch.

1.2 Project Resources

1.2.1 Hardware Resources

- **Development Computers:** High-performance PCs capable of running development tools and handling cloud services for AI integration.
- **Testing Devices:** Various devices such as Windows/Mac laptops, Android/iOS smartphones for cross-platform testing and responsiveness.

1.2.2 Software Resources

- **Web Development Frameworks:** React.js for front-end development, Node.js for back-end.
- **Databases:** FireBase for storing user data and system records.
- **AI Integration Tools:** OpenAI API for chatbot functionality.
- **Version Control System:** Git for source code management, with GitHub as the repository for collaboration.
- **Design Tools:** Figma or Adobe XD for UI/UX design.
- **Cloud Hosting:** AWS or Google Cloud for server hosting, database management, and AI services integration.

1.3 Project Organization

Roles:

- **Front-End Developer:** Responsible for developing the user interface, ensuring responsiveness, and integrating with the backend.
    - Patrick King & Nic Ortiz
- **Back-End Developer:** Focuses on server-side logic, database management, API development, and integrating AI chatbot functionalities.
    - Sam Prosser & Thomas Powell
- **UI/UX Designer:** Designs the interface and ensures a user-friendly experience across all features.
    - Maddie McDowell
- **Project Manager:** Ensures timely completion of tasks, coordination among team members, and maintaining the overall project timeline.
    - Patrick King & Maddie McDowell

Collaboration:

- Weekly team meetings will be held to review progress, discuss blockers, and set short-term goals. The project manager will facilitate these meetings and ensure alignment on priorities.

1.4 Project Schedule

1.4.1 PERT / GANTT Chart

A GANTT chart will be created using project management software such as Microsoft Project or Jira, to visualize the project timeline, highlighting key milestones, dependencies, and deadlines.

1.4.2 Task / Resource Table (Note: Some Overlap)

Task	Resource	Duration	Dependencies
Platform Design Document (PDD)	Entire team	2 Weeks	None
Front-End Development	Front-End Developer	2 Weeks	PDD
Back-End Development	Back-End Developer	2 Weeks	PDD
Career Questionnaire	Entire team	1 Weeks	PDD

Resume Analysis Tool	Back-End Developer	2 Weeks	Back-End Development
Cover Letter Generator	Back-End Developer	1 Weeks	Resume Analysis Tool
Job Matching Feature	Back-End Developer	2 Weeks	Front-End, Back-End
Learning Resources	Entire team	1 Weeks	Front-End, Back-End
Networking Tools	Entire team	1 Weeks	Front-End, Back-End
Alpha Version	Entire team	1 Weeks	Resume, Cover Letter Tools
Beta Version	Entire team	1 Weeks	Alpha Version
Testing & Feedback	Entire team	1 Weeks	Beta Version
Final Adjustments	Entire team	1 Weeks	Testing & Feedback
Launch Preparation	Entire team	1 Weeks	Final Adjustments
Post-Launch Support	Entire team	Ongoing	Launch